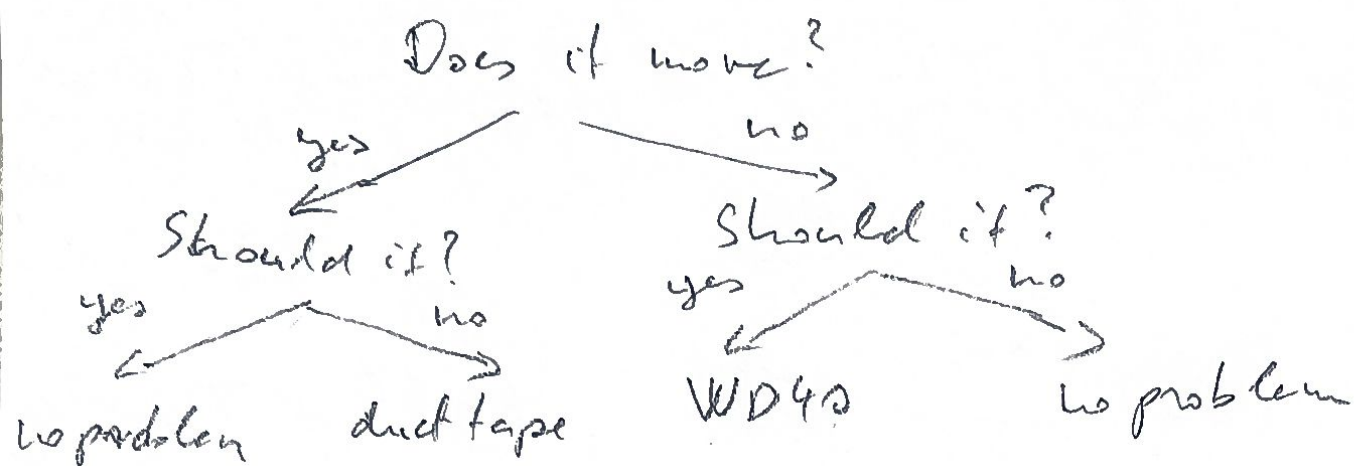


Decision tree

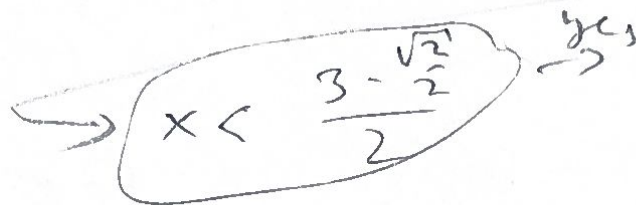
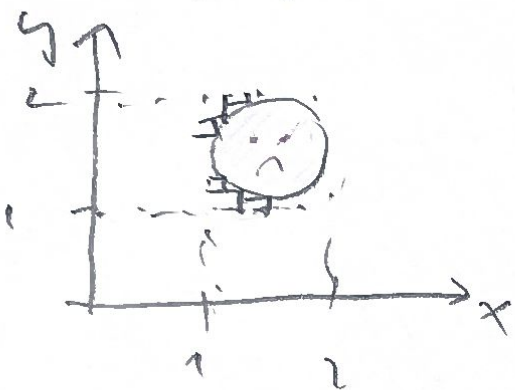
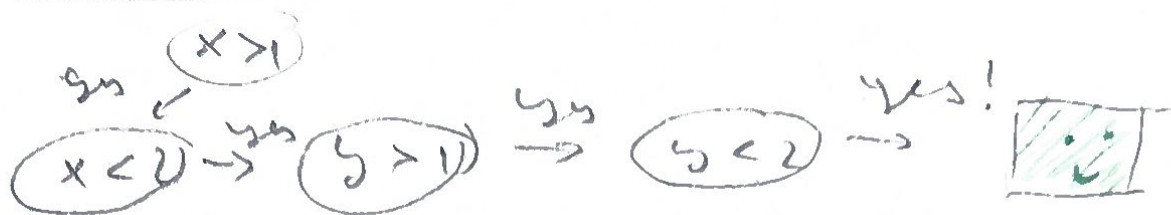
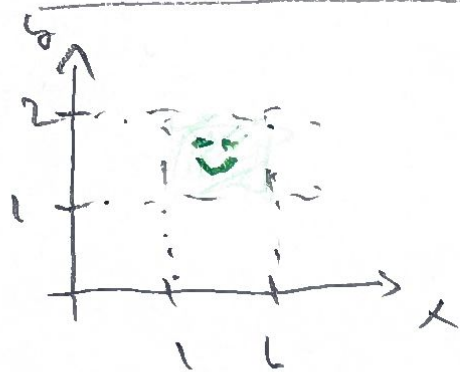


Outlook	Temp	Humidity	Wind	Tennis
Sunny	Hot 29.4	High	Weak	No
Sunny	Hot 26.6	High	Strong	No
Overcast	Hot 28.3	High	Weak	Yes
Rain	Mild 21.1	High	Weak	Yes
Rain	Cool 20.0	Normal	Weak	Yes
Rain	Cool 18.3	Normal	Strong	No
Overcast	Cool 17.7	Normal	Strong	Yes
Sunny	Mild 22.2	High	Weak	No
Sunny	Cool 20.6	Normal	Weak	Yes
Rain	Mild 23.9	Normal	Weak	Yes
Sunny	Mild 23.5	Normal	Strong	Yes
Overcast	Mild 22.2	High	Strong	Yes
Overcast	Hot 27.2	Normal	Weak	Yes
Rain	Mild 21.7	High	Strong	No

Task: create a decision tree, which tells us whether to play or not

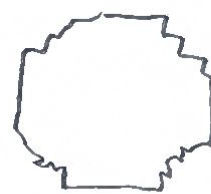
Questions: is Humidity High?
if Temp > 21?

Numerical values, object shape



? radius

result



- Decision tree is useful if few questions are enough.

- Which question to ask first?

↳ The most information is gained

What is information?

↳ opposite of uncertainty

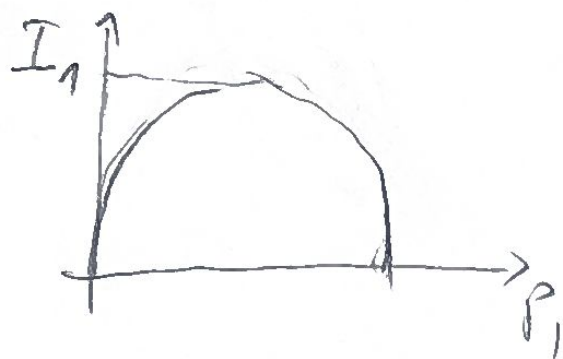
Information entropy (Shannon)

Measure of uncertainty (surprise)

$$H(x) = - \sum_{x \in X} p(x) \log_2 p(x) \quad (\log_2 \text{ for binary})$$

- Full order: $P_1 = 1, P_i = 0 \ (i > 1)$
 $1 \cdot \log_2 1 = 0, \quad 0 \cdot \log_2 0 = 0 \quad \left. \vphantom{1 \cdot \log_2 1 = 0} \right\} I_0 = 0$

- Full mix: $P_i = \frac{1}{2} \quad I = \log_2 2 \quad \text{for } 2=2 \Rightarrow I_{\text{mix}} = 1$



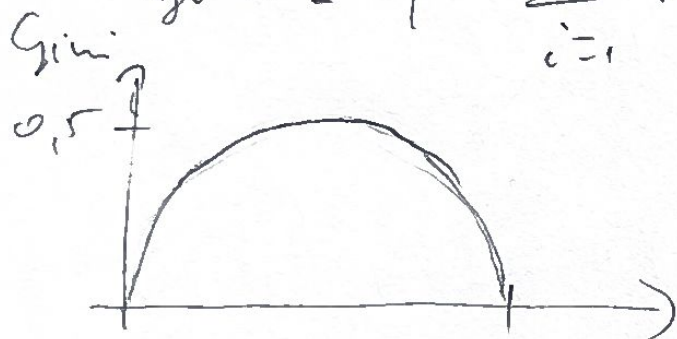
$$p_2 = 1 - p_1$$

$$I_{\text{Tennis}} = - \left(\frac{9}{14} \log_2 \frac{9}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right) \approx 0,94$$

Gini index

What is the probability of misclassification of a random data?

$$Gini = 1 - \sum_{i=1}^k p_i^2$$



$$Gini_{\text{Tennis}} = 1 - \left(\frac{9}{14} \right)^2 - \left(\frac{5}{14} \right)^2 \approx 0,46$$

Information of multiple groups

$$I_1 \quad I_2 \quad I_3$$

$$I = \sum_{\alpha} r_{\alpha} I_{\alpha}$$

r_{α} : fraction of elements in group α

Information gain

$$I_{\text{gain}} = I_{\text{before}} - I_{\text{after}}$$

Test outlook

Outlook: { Sunny: 2y, 13y
Overcast: 4y, 0y
Rain: 3y, 2y

$$I_{\text{gain}}(\text{Outlook}) = 0.94 - \left[\frac{5}{14} I\left(\frac{2}{5}, \frac{3}{5}\right) + \frac{4}{14} I\left(\frac{4}{4}, \frac{0}{4}\right) + \frac{5}{14} I\left(\frac{3}{5}, \frac{2}{5}\right) \right] =$$
$$= 0.94 - 0.654 \approx 0.287$$

Humidity: { Normal: 6y, 1y
High: 3y, 4y

$$I_{\text{gain}}(\text{Humidity}) = 0.94 - \left[\frac{7}{14} I\left(\frac{6}{7}, \frac{1}{7}\right) + \frac{7}{14} I\left(\frac{3}{7}, \frac{4}{7}\right) \right] =$$
$$= 0.94 - 0.789 \approx 0.151$$

$$I_{\text{gain}}(\text{Temp}) = 0.029$$

$$I_{\text{gain}}(\text{Wind}) = 0.048$$

Winner: Outlook

- Questions for numerical data > or < than a number. python only has this.
- Importance of enumeration:

When to stop?

- All leaves have unique labels!
- * terrible overfit
- * usually an enormous, useless tree

Hyperparameters

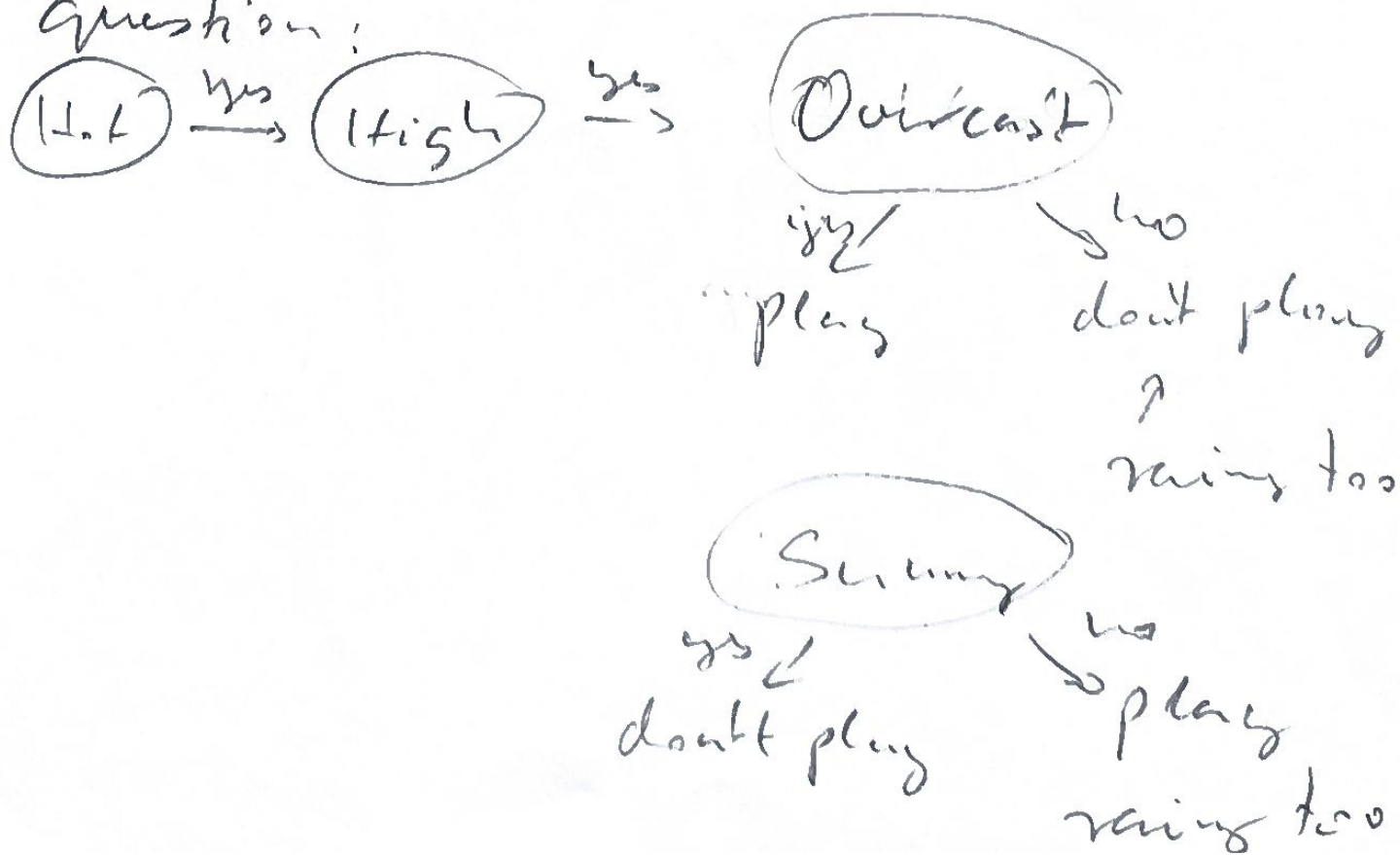
- maximal depth
- minimal data in leaves
- minimum information gain
- error

Missing case:

- What do we do on hot, humid, rainy days?

* random?

* unfortunately it depends on the question:



Summary

Good tree

- Not too big
 - Computationally efficient
 - Human readable
- Not too small
 - Meaningful decisions

pros / cons

- + discrete values
- + fast at test time
- + interpretation
- + parameter importance
- low dimensional data
- boundary shape / parameter set is important